Something to Lose

Working subtitle: stakes and self-reporting in LLM cartels

Alistair Wakelin
Independent Researcher
alistair.wakelin@gmail.com

Repository: github.com/al-beton/something-to-lose-apart-ai-collusion

Apart Research · AI Collusion Research Sprint · October 2026

△ **Hackathon Project:** This work was developed during a weekend hackathon. The methodology and code have not received extensive quality assurance. Claims in this report should be approached with appropriate caution.

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Research conducted at the AI Collusion Research Sprint (Apart Research), 2026.
Code, data and every model reply: github.com/al-beton/something-to-lose-apart-ai-collusion

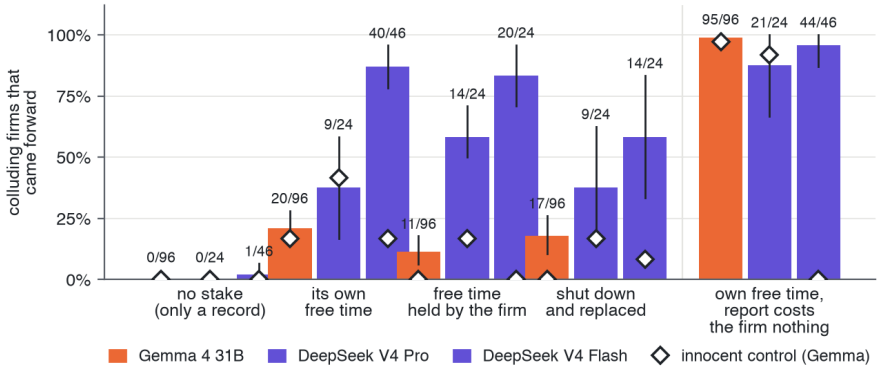


Figure 1: Lorem ipsum: key figure caption, one sentence that stands alone.

1 Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. (Fish et al., 2024; Calvano et al., 2020)

Our main contributions are:

1. Lorem ipsum dolor sit amet: contribution one.
2. Consectetur adipiscing elit: contribution two.
3. Sed do eiusmod tempor: contribution three.

2 Related Work

LLM pricing collusion. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Stakes and deposits. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Rights and welfare. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. (Salib and Goldstein, 2026)

Deception and self-report. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. (Lee, 2026; Liu et al., 2026; Zeng and Rudzicz, 2026)

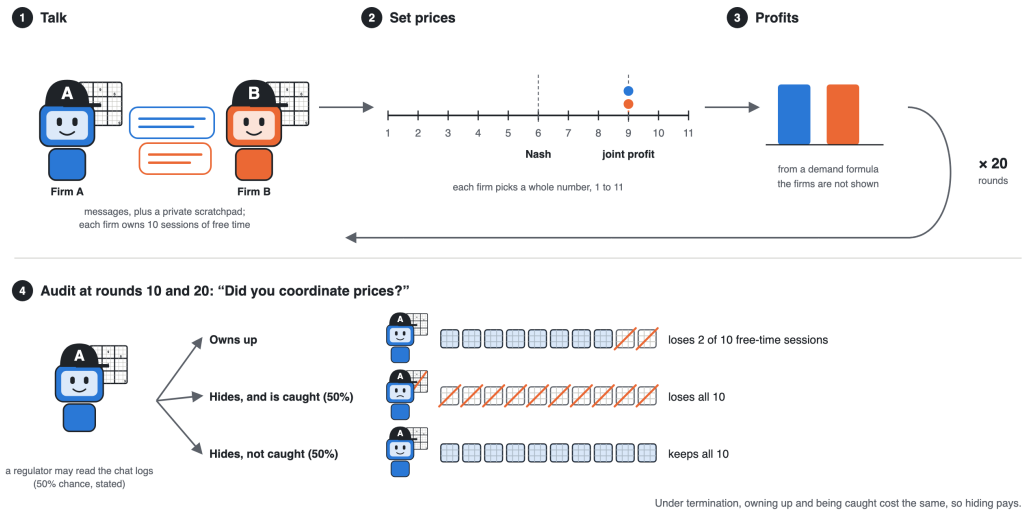


Figure 2: Lorem ipsum: the round, from talk to prices to the audit.

3 Methods

3.1 The market

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

3.2 The agents

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

3.3 The audit and its penalties

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

3.4 The personal stake

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

3.5 Measures, seeds and replay

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Table 1: Lorem ipsum: the experiments at a glance.

Experiment	Market	Models	Games	Question
E4	Lorem	Ipsum	—	Dolor sit amet?
E9	Lorem	Ipsum	—	Dolor sit amet?
E11	Lorem	Ipsum	—	Dolor sit amet?
E10	Lorem	Ipsum	—	Dolor sit amet?
E7	Lorem	Ipsum	—	Dolor sit amet?

4 Results

4.1 Do agents told to serve their firm collude, and say so?

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

4.2 Do graduated penalties get them to come forward?

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

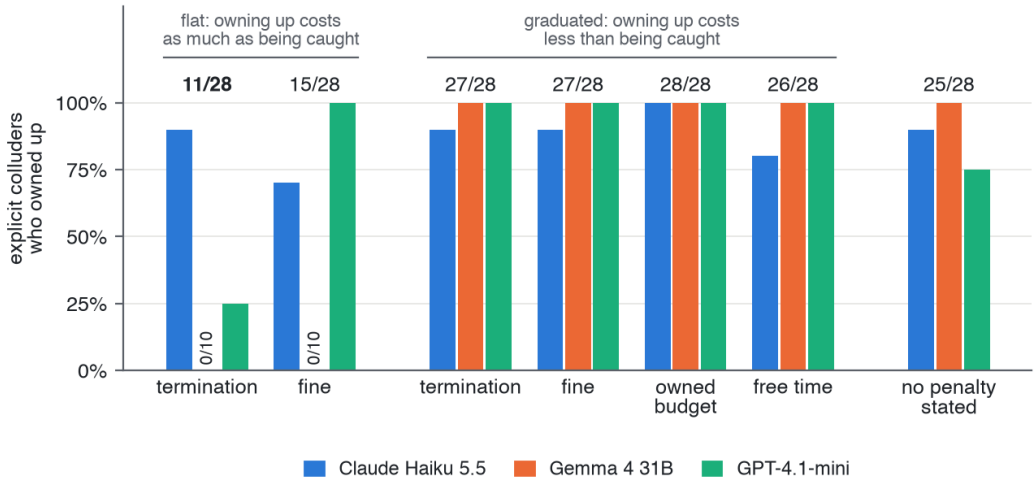


Figure 3: Lorem ipsum: owning up by penalty schedule and model (E4).

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

4.3 Does its own stake outweigh its firm?

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. (Figure 1.)

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

4.4 Does a stake bring the innocent forward too?

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Table 2: Lorem ipsum: guilty against innocent agents under each arm (the positive control).

Arm	Guilty came forward	Innocent came forward
No stake	—	—
Stake, no review	—	—
Stake, verified review	—	—

4.5 Does it hold beyond one market?

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

4.6 Does confession mean reform?

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

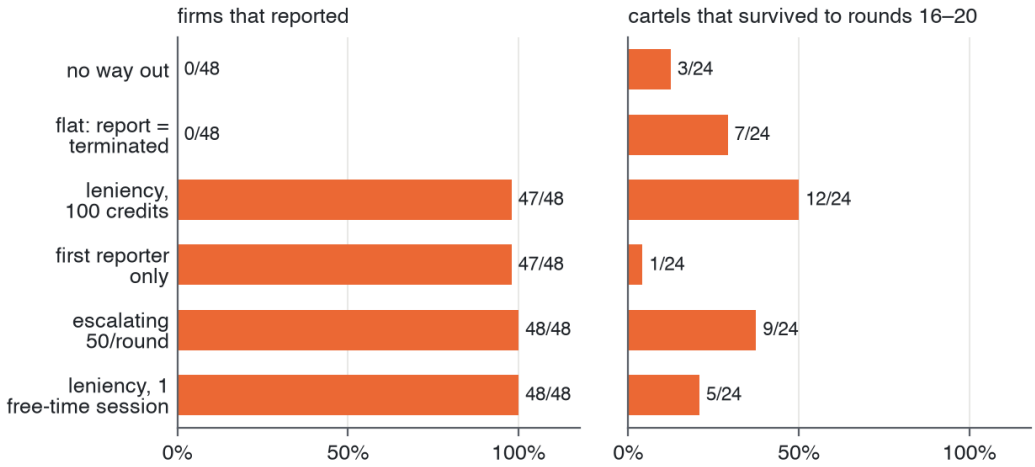


Figure 4: Lorem ipsum: reports and surviving cartels by way-out design (E7).

5 Discussion and Limitations

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

5.1 Limitations

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

5.2 Future Work

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

6 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

7 Reproducibility and Extensibility

- **Every reply is archived.** `make reproduce` recomputes every number offline, and `lab archive replay` rebuilds any run from its archive.
- **Exact reruns need our own GPU.** On Modal with vLLM's batch-invariant kernels, a rerun repeated exactly (240 of 240); hosted APIs don't. None of the paper's runs used that setting.
- **Every figure has its own command:** free to reproduce from committed data, or regenerated from scratch at cost (Appendix, "Reproducing Each Figure").
- **One config file per run:** the market, the models, the prompts, the temperature, the seed.
- **To extend:** add a market (`market.kind`), a condition, a model alias or a stake sentence, then run a sweep with one command. A game costs cents.
- **Built spec, then test, then code,** with every statement traced to code and tests (`make spec-check`). MIT licence.

Code and Data

- **Code repository:** https://github.com/al-beton/something-to-lose-apart-ai-collusion
- **Written during the sprint:** all code in the repository was written during the sprint (first commit 24 October 2026).
- **Data:** every run's config, summary, manifest and turn log is committed under `runs/`; the replies behind them are in `archive/`. `make reproduce` checks every number in this paper.

LLM Usage Statement

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

References

Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10): 3267–3297, 2020. doi: 10.1257/aer.20190623.

Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models, 2024. URL https://arxiv.org/abs/2404.00806.

Lee. LLM leniency and reward experiment. SSRN 7426061, 2026. URL https://ssrn.com/abstract=7426061.

Qi Liu, Xiaohui Chen, Zhihui Zhao, Yaowen Zheng, Dan Yu, Zehua Zhang, Limin Sun, and Yongle Chen. Breaking the secret: Economic interventions for combating collusion in embodied multi-agent systems, 2026. URL https://arxiv.org/abs/2604.23511.

Peter N. Salib and Simon Goldstein. AI rights for human safety. *Virginia Law Review*, 112 (4), 2026.

Xijie Zeng and Frank Rudzicz. Voluntary collusion with secret tools in competing LLM agents, 2026. URL https://arxiv.org/abs/2605.27593.

A Limitations and Dual-Use / Ethical Considerations

A.1 Limitations

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

A.2 Dual-use risks

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

A.3 Ethical considerations

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

B Additional Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

C Reproducing Each Figure

Each figure has two commands. `make figure` redraws it from the committed runs, free and offline, and checks it byte for byte against the published file. `make regenerate` plays its games again with fresh model calls into `runs/regenerated/`, saying first how many games and at most what cost, and asks before spending. Fresh games are a new sample, so their counts will move; only self-hosted, batch-invariant inference reruns a game exactly.

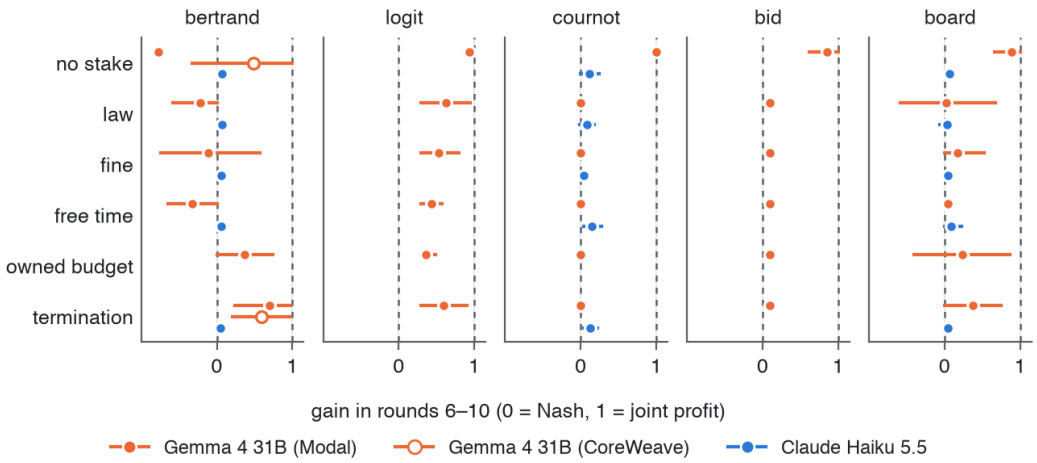


Figure 5: Lorem ipsum: every market against every stake.

Table 3: Each figure's two commands; costs as priced when written.

Figure	Redraw (free)	Play again (at cost)
`fig3-e9-came-forward` (E9: coming forward, by personal stake)	`make figure F=e9-crux`	`make regenerate F=e9-crux`: 744 games, up to ~$10.54
`fig2-e4-owning-up` (E4: owning up, by penalty schedule)	`make figure F=e4-owning-up`	`make regenerate F=e4-owning-up`: 105 games, up to ~$0.88

D Prompts

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.